

# Sai Karthik Chalamalasetty

Senior AI-ML Engineer | GenAI & RAG | NLP | MLOps | Python | PyTorch | AWS | Azure

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## PROFESSIONAL SUMMARY

- Senior AI/ML Engineer with 9+ years of progressive experience across data science, machine learning engineering, NLP, predictive analytics, financial risk modeling, and enterprise AI solutions in healthcare and financial-services environments.
- Progressed from Python/SQL analytics and classical machine-learning prototyping into ownership of production-grade ML and GenAI systems spanning data ingestion, feature engineering, model development, API integration, containerized deployment, monitoring, and model lifecycle management.
- Designed and implemented end-to-end machine learning workflows covering data profiling, feature engineering, model training, hyperparameter optimization, validation, explainability, packaging, deployment, observability, and production support using Scikit-learn, XGBoost, LightGBM, PyTorch, and TensorFlow.
- Developed NLP and document-intelligence solutions using text preprocessing, transformer models, embeddings, semantic search, retrieval-augmented generation, and evaluation frameworks to improve information discovery and decision support from unstructured enterprise content.
- Built production-oriented predictive models for credit risk, fraud detection, financial analytics, customer behavior, and healthcare use cases, with emphasis on model reliability, error analysis, interpretability, and alignment with business-risk requirements.
- Implemented reproducible MLOps practices using MLflow, DVC, Apache Airflow, Docker, Kubernetes, CI/CD, Prometheus, Grafana, AWS, and Azure services to support repeatable model development, deployment, monitoring, and controlled production releases.
- Applied explainable-AI techniques including SHAP, LIME, feature attribution, and model-behavior analysis to improve transparency, investigate prediction drivers, and support risk, research, and business review processes.
- Partnered with product, engineering, analytics, research, risk, compliance, and business stakeholders to translate ambiguous requirements into scalable AI/ML solutions and move models from proof-of-concept through validation and production deployment.

## TECHNICAL SKILLS

1. **Programming Languages & Data Analysis:** Python, SQL, Pandas, NumPy, PySpark, Apache Spark
2. **Machine Learning:** Scikit-learn, XGBoost, LightGBM, feature engineering, classification, regression, model validation, hyperparameter tuning
3. **Deep Learning & NLP:** PyTorch, TensorFlow, Hugging Face, BERT, spaCy, NLTK, embeddings, semantic search, document classification
4. **GenAI:** RAG, LangChain, FAISS, prompt orchestration, retrieval evaluation, response grounding
5. **MLOps & Deployment:** MLflow, DVC, Apache Airflow, FastAPI, Flask, Docker, Kubernetes, KServe, TensorFlow Serving, TorchServe
6. **Cloud & Data Platforms:** AWS SageMaker, EC2, S3, Step Functions, Azure ML, Azure Monitor, Databricks, Delta Lake, MongoDB, MySQL, PostgreSQL
7. **DevOps & Observability:** Git, GitHub Actions, Jenkins, Terraform, Prometheus, Grafana, model/service monitoring
8. **Explainability:** SHAP, LIME, feature attribution, error analysis, model transparency

## WORK EXPERIENCE

Elevance health, San Mateo, California, US | AI/ML Engineer

Dec 2025 – Present

- **Project: Enterprise Healthcare AI & Document Intelligence Platform**
- Designed an end-to-end AI workflow for healthcare and benefit-related content, beginning with document ingestion and validation, followed by parsing, normalization, metadata extraction, chunking, embedding generation, retrieval, response generation, and evaluation before application integration.
- Built retrieval-augmented generation workflows using transformer/embedding models, FAISS-based semantic retrieval, and LangChain orchestration to surface grounded information from unstructured enterprise documents while preserving traceability to source content.
- Developed Python and FastAPI services for model and retrieval inference, defined Pydantic request/response contracts, and collaborated with application teams to integrate AI capabilities into member- and associate-facing workflows.
- Implemented evaluation checks for retrieval relevance, answer grounding, response consistency, and failure modes; used test datasets and error analysis to refine chunking, retrieval parameters, prompts, and model-selection decisions.
- Applied privacy-aware data handling, access controls, input validation, and responsible-AI review practices for sensitive healthcare information, with emphasis on explainability, robustness, security, and appropriate human oversight.
- Containerized AI services with Docker and supported Kubernetes-based deployment patterns; established CI/CD checks and environment-specific configuration for repeatable promotion across development, test, and production.
- Instrumented application and model-service telemetry using Azure Monitor and operational dashboards to track latency, failures, resource utilization, and service health; investigated production issues using logs and trace-level diagnostics.

- Maintained experiment and model artifacts, documented architecture and evaluation decisions, and partnered with product, engineering, analytics, and compliance stakeholders to move prototypes into production-ready AI capabilities.

**Environment:** Python, PyTorch, Hugging Face, LangChain, FAISS, RAG, FastAPI, Pydantic, Docker, Kubernetes, Azure ML, Azure Monitor, MLflow, SQL, GitHub Actions, Terraform

## **TransUnion, Illinois, United States | ML Engineer**

**Jul 2024 – Nov 2025**

### **Project: Credit Risk & Fraud Decisioning ML Platform**

- Developed production-oriented machine learning workflows for credit-risk scoring, fraud analysis, and predictive decisioning using credit attributes, account history, transaction signals, and alternative data sources.
- Built reusable feature pipelines in Python, SQL, and PySpark to validate schemas, handle missing values and outliers, encode categorical attributes, aggregate behavioral signals, and generate point-in-time model inputs for repeatable training and inference.
- Trained and compared classification models using Scikit-learn, XGBoost, and deep-learning frameworks; tuned hyperparameters and evaluated precision, recall, F1, ROC-AUC, calibration, and error segments to select models appropriate for business-risk tradeoffs.
- Applied SHAP and LIME to explain model predictions, identify influential risk factors, investigate unexpected decisions, and support transparent review of model behavior in credit and fraud workflows.
- Managed experiments, datasets, hyperparameters, and model versions using MLflow and DVC, enabling reproducible comparisons and controlled promotion of candidate models through development and validation cycles.
- Packaged model artifacts behind inference services, containerized workloads with Docker, and supported Kubernetes-based model serving with health checks, configuration management, and scalable execution patterns.
- Automated recurring data preparation and model workflows with Apache Airflow and integrated GitHub Actions/Jenkins checks for testing, image builds, and controlled deployment activities.
- Implemented Prometheus/Grafana monitoring for model-service availability, latency, resource consumption, and operational failures; profiled PySpark and training workloads to improve execution efficiency and production reliability.
- Collaborated with data engineering, application engineering, analytics, and risk stakeholders to translate decisioning requirements into model features, validation criteria, deployment interfaces, and production support procedures

**Environment:** Python, PySpark, Scikit-learn, XGBoost, PyTorch, TensorFlow, MLflow, DVC, Airflow, Docker, Kubernetes, AWS SageMaker, EC2, S3, SQL, MongoDB, SHAP, LIME, Prometheus, Grafana, GitHub Actions, Jenkins, Terraform

## **Vanguard, Valley Forge, PA | Data Scientist (ML)**

**May 2022 – Jun 2024**

### **Project: Financial Research NLP & Quantitative ML Analytics**

- Developed NLP and machine-learning workflows to analyze earnings transcripts, research reports, market information, and portfolio-related datasets, converting unstructured and structured financial data into model-ready features for research and analytical use cases.
- Built text-processing pipelines for normalization, tokenization, embedding generation, sentiment analysis, topic extraction, and document classification; evaluated transformer-based approaches where appropriate for the 2022-2024 technology stack.
- Prepared point-in-time datasets from historical market signals, asset characteristics, performance records, and research indicators, with validation controls designed to reduce leakage and preserve consistency between training and evaluation periods.
- Developed and evaluated predictive models using cross-validation, walk-forward/backtesting workflows, error analysis, and stability checks to understand model behavior across market regimes and identify opportunities for refinement.
- Applied SHAP and LIME to examine feature contributions and prediction drivers, improving interpretability for research users and supporting review of model outputs before analytical adoption.
- Managed experiments and model artifacts using MLflow and DVC, recording dataset versions, training parameters, evaluation results, and candidate model changes to support reproducible development.
- Automated data preparation and model execution with Apache Airflow and AWS Step Functions; organized training data and model artifacts in S3 and executed experimentation/deployment workloads using EC2 and SageMaker.
- Packaged validated models for inference and supported integration through TensorFlow Serving and TorchServe, performing prediction testing and environment validation before analytical use.
- Partnered with research and business teams to review findings, investigate unexpected model behavior, document methodology, and refine analytical approaches based on domain feedback.

**Environment:** Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, Airflow, MLflow, DVC, AWS SageMaker, Step Functions, S3, EC2, TensorFlow Serving, TorchServe, SHAP, LIME

## **Cognizant, Bengaluru, India | Data Scientist**

**Sep 2018 – Dec 2021**

### **Project: Customer Interaction NLP & Intelligent Service Analytics**

- Developed NLP and predictive-analytics components for banking and insurance service workflows, progressing from classical text features and supervised models into more advanced language-model approaches as the technology matured.
- Prepared labeled customer-interaction datasets through cleansing, normalization, tokenization, label validation, exploratory analysis, and train/validation splitting; documented data-quality issues and worked with analysts to improve training-set reliability.
- Built intent-classification and sentiment-analysis models using TF-IDF/feature engineering, Scikit-learn, XGBoost/LightGBM, spaCy, and NLTK; evaluated precision, recall, F1, confusion matrices, and misclassification patterns to guide iterative model improvement.

- Developed multilingual preprocessing workflows for communication data, including normalization, tokenization, vectorization, and language-specific cleansing to create consistent inputs for downstream classifiers.
- Prototyped transformer/BERT-based text classification during the later part of the role for suitable document and interaction use cases, comparing results with established baselines before recommending adoption.
- Created Python evaluation utilities and reusable data-preparation modules, refactoring experimental notebooks into maintainable components with version control and peer-review practices.
- Exposed validated ML components through Flask APIs and supported integration with downstream applications; used AWS EC2/S3 for datasets and analytical workloads and MySQL/PostgreSQL for structured data extraction.
- Worked with developers, analysts, senior data scientists, and business stakeholders to translate requirements into model features, acceptance criteria, and technical implementation tasks while strengthening production-delivery skills.

**Environment:** Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, spaCy, NLTK, BERT (later-stage prototypes), Flask, AWS EC2, AWS S3, MySQL, PostgreSQL, Git, GitHub

**Luxoft India Pvt. Ltd., Bangalore, India | Associate Data Scientist**

**Apr 2017 – Aug 2018**

- **Project: Customer Behavior, Churn & Experimentation Analytics**
- Supported senior data scientists by extracting and preparing structured datasets with Python and SQL, performing exploratory data analysis, validating source completeness, and documenting patterns, anomalies, and data-quality issues.
- Built baseline machine-learning prototypes for customer behavior and churn analysis using feature preparation, train/test splitting, cross-validation, and classical classification techniques; compared model results and documented limitations for senior review.
- Performed statistical analysis for A/B testing and behavioral experiments by preparing cohorts, calculating comparison metrics, checking data assumptions, and summarizing findings for analysts and project leads.
- Created dashboards and visualizations for customer segments, engagement trends, and operational metrics, helping business users understand patterns before more advanced modeling was undertaken.
- Refactored EDA notebooks into reusable Python modules with validation checks, clearer function boundaries, and Git-based version control; incorporated feedback from peer code reviews to improve coding standards and maintainability.
- Maintained data dictionaries, transformation logic, model-evaluation notes, and analytical documentation to support repeatability and knowledge transfer across project teams.

**Environment:** Python, Pandas, NumPy, Scikit-learn, SQL, AWS S3, MySQL, PostgreSQL, Git, GitHub

**EDUCATION**

**SRM University (Sri Ramaswamy Memorial University)**

**Jul 2014 – Jun 2018**

Bachelor of Technology in Computer Engineering